

Wethum Lansakara

Computer Science & AI Undergraduate

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence at NIBM, with hands-on experience in machine learning, model evaluation, and AI-powered software development. Skilled in Python, data preprocessing, exploratory data analysis, classification, regression, and database technologies. Seeking an internship to apply technical and analytical skills to real-world problems while gaining industry experience.

ACADEMIC PROJECTS

StudyPilot AI – AI-Powered Study Platform

- Developed an AI-powered study platform with a FastAPI backend and PostgreSQL database hosted on Neon.
- Integrated OpenRouter API for AI-powered question answering and content summarization.
- Implemented REST API-based frontend/backend communication and deployed the application using Vercel.

Mobile Price Range Classification – Machine Learning

- Developed a multi-class classification system to predict mobile phone price ranges based on technical specifications.
- Performed data cleaning, outlier analysis, exploratory data analysis (EDA), correlation analysis, and feature relationship visualization.
- Compared KNN, Logistic Regression, SVM, Decision Tree, and Naive Bayes classification models.
- Achieved the highest accuracy of 95.97% with Logistic Regression, outperforming the other evaluated models.

Autonomous Scarecrow Robot – IoT & Robotics

- Developed an autonomous agricultural monitoring robot using Arduino UNO and ESP32.
- Implemented PID line following, PIR motion detection, servo scanning, and ultrasonic obstacle detection.
- Integrated Firebase and a React/Vite dashboard for real-time monitoring and remote control.

Sri Lankan Rainfall Prediction – Machine Learning

- Analyzed a real-world Sri Lankan rainfall dataset containing 55,000+ records across multiple administrative regions.
- Performed data cleaning, feature engineering, exploratory data analysis, statistical analysis, and hypothesis testing.
- Developed and compared Linear Regression and Random Forest Regression models for rainfall prediction.
- Achieved the best performance with Random Forest (R^2 : 69.3%, RMSE: 33.25, MAE: 21.14).

Smart Study Planner – AI-Powered Study Scheduling

- Developed an intelligent study planner using Python, Flask, SWI-Prolog, and MySQL.
- Used Prolog-based reasoning to prioritize and allocate study time based on factors such as exam proximity, difficulty, and progress.
- Built a web interface for managing subjects and personalized study schedules.

EDUCATION

HND in Computer Science with Artificial Intelligence

2026 – Expected Dec 2026

National Institute of Business Management

Diploma in Computer Science with Artificial Intelligence

2025 – 2026

National Institute of Business Management

Foundation Degree in Computer Science with Artificial Intelligence

2024

National Institute of Business Management

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript

AI & Machine Learning: Scikit-learn, Pandas, NumPy, OpenCV, OpenRouter API, LLM Integration, Classification, Regression, EDA, Feature Engineering

Web Development: HTML, JavaScript, CSS, FastAPI, Flask, React

Database Management: MySQL, Firebase, MongoDB, PostgreSQL, Neon

Tools & Platforms: GitHub, VS Code, Jupyter Notebook, Git

Other: OOP, Agile Concepts, Model Evaluation

SOFT SKILLS

Problem Solving, Team Collaboration, Time Management, Continuous Learning, Communication Skills

CERTIFICATION

Python for Beginners

University of Moratuwa

Web Hacking for Beginners

Udemy

Machine Learning Online Course

Udemy

Critical Thinking in the AI Era

HP LIFE

Java Mastery: Beginner's Guide and Full-Stack AI Projects

Udemy

LANGUAGES

English (Intermediate), Sinhala (Fluent)

LINKS

GitHub: <https://github.com/wethumlansakara>

LinkedIn: <https://www.linkedin.com/in/wethum-lansakara>